

Eine Woche, ein Beispiel

5.25 cyclic coverings

Main Ref: [Barth04, I.17]

https://archive.ymsc.tsinghua.edu.cn/pacm_download/27/8789-Introduction-cyclic-cover.pdf

[Barth04]: Barth, Wolf P., Klaus Hulek, Chris A. M. Peters and Antonius Van De Ven. Compact Complex Surfaces.

Setting $Y/\mathbb{C}$ conn cplx mfld (or sm integral variety)
 $B \in \text{Div}(Y)$ effective
 $\mathcal{L} \in \text{Pic}(Y)$ with $\mathcal{L}^{\otimes m} \cong \mathcal{O}_Y(B)$ $m \in \mathbb{N}_{\geq 1}$
 $s \leftrightarrow 1$

$$\begin{array}{c} \text{Spec}(\mathcal{O}_Y \oplus \mathcal{L}^{-1} \oplus \dots \oplus \mathcal{L}^{-m+1}) \\ \parallel \\ \rightsquigarrow \mathcal{L} \triangleq \text{Spec}(\text{Sym } \mathcal{L}^{-1}) \supset \text{Spec}(\text{Sym } \mathcal{L}^{-1} / \langle s \rangle) \triangleq X \\ \downarrow p \quad \swarrow f \\ Y = \text{Spec}(\mathcal{O}_Y) \end{array}$$

Define (X, Y, f) as the m -cyclic covering of Y branched along B ,
determined by $\mathcal{L}$.
Rmk. X has at most singularities over singular pts of B .

Denote $B_i := f^{-1}(B) \in \text{Div}(X)$, then $f^*B = mB_i$

$$\begin{array}{ccccccc} X & \xrightarrow{\mathcal{O}_X(B_i)} & \mathcal{O}_X(mB_i) & \xrightarrow{\omega_X \otimes f^* \mathcal{L}^{m-1}} & \mathcal{O}_X & \xrightarrow{\quad} & \mathcal{O}_X(B_i) \\ & & & \Omega_X^m((1-m)B_i) & & & \\ f \downarrow & \xrightarrow{\mathcal{L}} & \mathcal{O}_Y(B) & \xrightarrow{\omega_Y} & \bigoplus_{j=0}^{m-1} \mathcal{L}^{-j} & \xrightarrow{\quad} & \bigoplus_{j=-1}^{m-2} \mathcal{L}^{-j} \\ Y & & & & & & \end{array}$$

pullback pushforward

L : the total space of the l.b. $\mathcal{L}$.

$$\begin{array}{ccc} X & \xrightarrow{l_X} & L \\ f \searrow & & \swarrow f_L \\ \{s=0\} = B & \subset & Y \end{array}$$

$$X = \{(y, t) \in L \mid t^m = f_L^* s\}$$

$$\Rightarrow \mathcal{O}_L(X) = f_L^*(\mathcal{O}_Y(B))$$

$$\begin{array}{l} t \in H^0(f_L^* \mathcal{L}) \\ s \in H^0(\mathcal{O}_Y(B)) \end{array}$$

E.g. For a hyperelliptic curve C with $f: C \xrightarrow{2:1} \mathbb{P}^1$, $g(C)=g$.

$$B_1 := \{\text{Weierstrass pts of } C\} = \{p_1, \dots, p_{2g+2}\}$$

$$B := f(B_1)$$

We define

$$\mathcal{L}_0 := f^* \mathcal{O}_{\mathbb{P}^1}(1) \in g_2,$$

then

$$\omega_C = f^* \mathcal{O}_{\mathbb{P}^1}(g-1) = \mathcal{L}_0^{g-1},$$

and

	ω_C	$\mathcal{O}_C(B_1)$	$\mathcal{O}_C(2B_1)$	$\omega_C \otimes \pi^* \mathcal{L}$		
	$\parallel$	$\parallel$	$\parallel$	$\parallel$		
C	$\mathcal{L}_0^{g-1}$	$\mathcal{L}_0^{g+1}$	$\mathcal{L}_0^{2g+2}$	$\mathcal{L}_0^{-2}$	$\mathcal{O}_C$	$\mathcal{O}_C(B_1)$
$f \downarrow$	$\mathcal{O}(1)$	$\mathcal{O}(g+1)$	$\mathcal{O}(2g+2)$	$\mathcal{O}(-2)$	$\mathcal{O} \oplus \mathcal{O}(-g-1)$	$\mathcal{O}(g+1) \oplus \mathcal{O}$
$\mathbb{P}^1$	$\mathcal{L}$	$\mathcal{O}_{\mathbb{P}^1}(B)$	$\omega_{\mathbb{P}^1}$		$\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{L}^{-1}$	$\mathcal{L} \oplus \mathcal{O}_{\mathbb{P}^1}$

pullback

pushforward

$$\text{div } \frac{y}{\prod_{i=g+2}^{2g+2} (x-p_i)} = \sum_{i=1}^{g+1} p_i - \sum_{i=g+2}^{2g+2} p_i$$

E.g. For a degree m cyclic cover $f: X \rightarrow \mathbb{P}^n$ ramified over a hypersurface $S \in \mathbb{P}^n$ of degree d , $m|d$

$$\begin{aligned} \mathcal{L}^m &= \mathcal{O}_Y(B) = \mathcal{O}_{\mathbb{P}^n}(S) = \mathcal{O}_{\mathbb{P}^n}(d) \Rightarrow \mathcal{L} = \mathcal{O}_{\mathbb{P}^n}\left(\frac{d}{m}\right) \\ \mathcal{O}_X(S) &\stackrel{\text{def}}{=} \mathcal{O}_X(f^{-1}(S)) = f^*\mathcal{L} = f^*\mathcal{O}_{\mathbb{P}^n}\left(\frac{d}{m}\right) \\ \mathcal{O}_L(X) &= f_L^*(\mathcal{O}_Y(B)) = f_L^*\mathcal{O}_{\mathbb{P}^n}(d) \end{aligned}$$

X	$\mathcal{O}_X(S) \quad \mathcal{O}_X(mS) \quad \omega_X \otimes f^*\mathcal{L}^{m-1}$ $\omega_X((1-m)S)$	$\mathcal{O}_X$	$\mathcal{O}_X(S)$
$f \downarrow$	$\mathcal{O}_{\mathbb{P}^n}\left(\frac{d}{m}\right) \quad \mathcal{O}_{\mathbb{P}^n}(d)$	$\omega_{\mathbb{P}^n} = \mathcal{O}_{\mathbb{P}^n}(-n-1)$	$\bigoplus_{j=0}^{m-1} \mathcal{O}_{\mathbb{P}^n}\left(-\frac{j d}{m}\right) \quad \bigoplus_{j=1}^{m-2} \mathcal{O}_{\mathbb{P}^n}\left(-\frac{j d}{m}\right)$
Y			
	pullback		pushforward

$$\begin{aligned} \omega_X &= \omega_X((1-m)S) \otimes \mathcal{O}_X(S)^{\otimes(m-1)} \\ &\cong f^*\left(\omega_{\mathbb{P}^n} \otimes \mathcal{O}_{\mathbb{P}^n}\left((m-1)\frac{d}{m}\right)\right) \\ &\cong f^*\mathcal{O}_{\mathbb{P}^n}\left(-n-1 + (m-1)\frac{d}{m}\right) \end{aligned}$$